



NIELS BOHR LIBRARY & ARCHIVES / TRANSCRIPT

ONE PHYSICS ELLIPSE, COLLEGE PARK, MD 20740 • (301)-209-3177 • NBL@AIP.ORG

P. A. M. Dirac - Session III

May 7, 1963

Interviewed by: Thomas S. Kuhn

Location: Dirac's home, Cambridge, England

Transcript version date: December 18, 2024

DOI: <https://doi.org/10.1063/nbla.ochm.ospg>

Abstract:

Part of the Archives for the History of Quantum Physics oral history collection, which includes tapes and transcripts of oral history interviews conducted with circa 100 atomic and quantum physicists. Subjects discuss their family backgrounds, how they became interested in physics, their educations, people who influenced them, their careers including social influences on the conditions of research, and the state of atomic, nuclear, and quantum physics during the period in which they worked. Discussions of scientific matters relate to work that was done between approximately 1900 and 1930, with an emphasis on the discovery and interpretations of quantum mechanics in the 1920s. Also prominently mentioned are: Niels Henrik David Bohr, Max Born, Boyland, Louis de Broglie, Johannes Martinus Burgers, Paul Ehrenfest, Ralph Fowler, Peter Fraser, Werner Heisenberg, Ernst Pascual Jordan, Cornelius Lanczos, Edward Arthur Milne, Wolfgang Pauli, David Robertson, Ernest Rutherford, Erwin Schrödinger, John Joseph Thomson, Hermann Weyl; University of Cambridge, Delta Squared V Club, Kapitsa Club, Kbenhavns~Universitet, Merchant Venturer's School in Bristol, University of Bristol, and Universität Göttingen.

Usage Information and Disclaimer:

This transcript, or substantial portions thereof, may not be redistributed by any means except with the written permission of the American Institute of Physics.

AIP oral history transcripts are based on audio recordings, and the interviewer and interviewee have generally had the opportunity to review and edit a transcript, such that it may depart in places from the original conversation. If you have questions about the interview, or you wish to consult additional materials we may have that are relevant to it, please contact us at nbl@aip.org for further information.

Please bear in mind that this oral history was not intended as a literary product and therefore may not read as clearly as a more fully edited source. Also, memories are not fully reliable guides to past events. Statements made within the interview represent a subjective appreciation of events as perceived at the time of the interview, and factual statements should be corroborated by other sources whenever possible or be clearly cited as a recollection.

Kuhn:

There is one thing which came up yesterday in which I understood you somewhat differently from our first talk in Princeton. That was this issue of quaternion's which may or may not be important, but which I'd like to pin down. When we talked in Princeton I understood you to say that you had learned something about them from reading Thomson and Tait while you were at Bristol.

Dirac:

I read some book; I might very well have the wrong title, I don't know. It was a big thick book.

Kuhn:

And it was devoted entirely to quaternion's?

Dirac:

Entirely, yes.

Kuhn:

And you think you will have read that book quite early? The reason I ask that is that John Wheeler once sat down to dinner at M.I.T. next to Egon Orowan who mentioned quaternion's and [who] said that he had once given you Tait's book on quaternion's; It's a book I don't know, but it's a likely book. That of course would be very much later.

Dirac:

It was probably when I was an engineering student that I read this about quaternion's.

Kuhn:

You think you had a whole book?

Dirac:

I didn't read the whole book through. I just read parts here and there.

Kuhn:

If it was a whole book on quaternion's, it would have to be something other than Thomson and Tait, I think. It could well have been Tait's Quaternion's.

Dirac:

That might be it, yes.

Kuhn:

Do you remember a did you use them at all in working out problems?

Dirac:

No, I did not use them.

Kuhn:

It was simply a question of knowing that they existed and being somewhat interested.

Dirac:

I found them interesting

Kuhn:

But you don't think you'd have used them in working out problems or subject matter for your own benefit?

Dirac:

No. I felt they ought to have an important application; I still feel that way, that they ought to have a more important application than they do at present.

Kuhn:

Do you think that given the relatively greater knowledge now and facility in utilizing tensors, general operator techniques, and so on, that the quaternion as something distinct from this would still have that sort of function?

Dirac:

Yes, yes I do. It is the most general algebra with division, and with associative

multiplication. That's what we need in quantum theory.

Kuhn:

Did you ever try working out quantum theory in a quaternion formulation?

Dirac:

I have, but I didn't get any new results from it. Other people have also been interested in that question.

Kuhn:

Does that go back very early, or is that fairly recent?

Dirac:

Well, off and on, a good many times. Quaternion's are something which I continually come back to.

Kuhn:

Did you try it at the time of the very early papers? And when you were doing a q-number algebra, for example, did you experiment with quaternion's?

Dirac:

I expect I did a few years later.

Kuhn:

Actually that again is out of order, but it raises a question which I had wanted to ask you. At the time of your thesis, both in the thesis itself, and about the same time In one paper on q-number algebra that's published in the Cambridge Philosophical Proceedings, you do work on the foundations of a q-number algebra.

Dirac:

Yes.

Kuhn:

Did you go on with that? You don't carry it very far at that me, and I see nothing later

that's an attempt, if you will, to behave as a mathematician with respect to this sort of algebra.

Dirac:

At that time I didn't properly realize that this was really exactly equivalent to matrix algebra, and later on when I realized that I thought it wasn't really a subject to study in itself... Mathematicians already know about matrix algebra, and this was really equivalent to that.

Kuhn:

This is taking matrix algebra in its most general form, to include continuous indices?

Dirac:

Yes, yes; operator algebra algebra of linear operators. You don't have to use continuous indices because you're working in Hilbert space, and you only use continuous indices when it is useful because of the physical meaning of the indices.

Kuhn:

When you say the mathematicians already knew all about these, in a sense I suppose they did, but there were a number of manipulative procedures and all sorts of mathematician's questions that might have been asked about existence, convergence and the whole question of what mathematicians know about the Delta function as a legitimate mathematical tool. In this sense, then, there was a good deal of mathematics used in the physics of these problems as it developed which the mathematicians were really not prepared to stand behind, so that work could have been done there.

Dirac:

The mathematicians wanted a higher standard of rigor than was necessary for the physicists. Well, after a few years Wintner wrote his book which gives, I suppose, practically all one needs to know so far as concerns the bounded matrices.

Kuhn:

That's a book I don't know.

Dirac:

I've got it here. [Goes to find the book.] It's not really adequate for quantum theory,

because it deals only with bounded matrices, but it's really a foundation of the subject. One ought to know all about the bounded matrices before one goes onto the unbounded ones. And that's all treated with complete rigor. I've often referred to that book when I wanted to know just what is certainly true with absolute rigor.

Kuhn:

Good. For the record I'm just going to read the title: it's A. Wintner, Spektraltheorie der Unendlichen Matrizen, Leipzig, 1929. No, I don't know this book at all.

Dirac:

The 'Unendlich' means an infinite number of rows and columns, but it's still a bounded matrix.

Kuhn:

Did you ever involve yourself with the mathematicians try to persuade the mathematicians to do more work on the more general sorts of matrices that quantum mechanics was involved with?

Dirac:

No, I didn't, no.

Kuhn:

Did any of the Cambridge people take this up? I can't remember. One could tell that by looking at the published papers.

Dirac:

So far as I know, there was remarkable little follow up from mathematicians - even on the whole in Germany, where the ties were closer and where some of the mathematics that had been handled, particularly by Hubert and his group, was very close to some of the techniques. I think the mathematicians mostly go their own way, and they're not much influenced by physics. Although the Schrodinger theory of the hydrogen atom was the first example that was discovered of a linear operator with both discrete and continuous Eigen values. You knew that, did you?

Kuhn:

Yes. There is one other question that I simply wanted to check on when we stopped

yesterday. I had raised with you the question of the likelihood that you were actually referring to de Brogue's formulation when you make the remark in the early paper that you can't utilize the notion of light quanta in thinking about the sorts of statistical problems you've been treating in the paper on detailed balancing.

Dirac:

I did know about de Brogue's work then; I had read some of it. Of course what I said was wrong, because I didn't then understand that one could have Bose statistic for particles with non-zero rest mass.

Kuhn:

I wonder whether at the point that paper was done, you knew Bose statistics at all?

Dirac:

No, no. That wasn't discovered until later.

Kuhn:

It's not much later; those two Einstein papers come out just at the end of 1924 and the beginning of '25.

Dirac:

Yes.

Kuhn:

But it wouldn't quite have been there. Did you see those papers immediately?

Dirac:

I think I did, yes, but I did not appreciate their importance.

Kuhn:

Was Fowler at all interested in them?

Dirac:

I think he was, yes.

Kuhn:

They fall right into his area of interest; on the other hand it's perfectly possible to say that this is nice mathematics, but just mad.

Dirac:

I had read Fermi's paper about the Fermi statistics and forgotten it completely.

Kuhn:

Had you?

Dirac:

Absolutely forgotten it, and when I wrote up my work on the antisymmetrical wave functions, I just didn't refer to it at all because I had completely forgotten it. Then Fermi wrote and told me, and I remembered that I had previously read about it.

Kuhn:

That's terribly interesting. Do you have an notion how that paper had seemed to you when you read it?

Dirac:

Well, I saw that it was the right statistics for electrons in an atom, but I didn't feel that it had wider applications. It was just a question of not appreciating a generality of the ideas.

Kuhn:

I'm very much interested both in the fact and in the way in which that sort of thing happens again and again I think.

Dirac:

It shows what a bad memory I have. If something doesn't strike me as being specially important, it's liable to slip out of my mind altogether.

Kuhn:

I take 'specially important' as fitting in to something that you're actively on at the time. Good. Well, then I'd like to go back more or less to the outline, and start by talking

about your coming to Cambridge. You lived in St. John's when you came here?

Dirac:

I lived in lodgings. The first year I was in lodgings; the second year I was in College. The third year I was in lodgings again.

Kuhn:

What determines that?

Dirac:

Well, there are too many students and not enough rooms in college, so that as far as possible they're put in a college. Those who hold scholarships are given precedence; they can always live in college. I had, an exhibition which didn't have the same standing as a scholarship. I think it was perhaps the general rule then for students to be living out the first year and in college the second year.

Kuhn:

What would then move you back out in the third year having gotten in in the first place?

Dirac:

I think there were just one in three living in college. Anyway Cunningham could tell you about these things because he was tutor for a very long time and dealt with these things. I met Cunningham when I first came to Cambridge for the purpose of the exam, in 1921. The only person I did meet I think.

Kuhn:

Living out, did you then live entirely by yourself? Did you eat in hall?

Dirac:

Yes, yes I did eat in hall every evening.

Kuhn:

Where did you work?

Dirac:

Sometimes in my lodgings, sometimes in libraries. My lodgings were often cold, and in cold weather I moved to a warm library. There were several libraries available. There were three: the Library of the Philosophical Society, the University Library, and the College Library.

Kuhn:

Does the Cavendish have a science library of its own in addition to these?

Dirac:

Oh yes, that would make four libraries available. Yes, the Cavendish had a little library of its own. And of course it wasn't nearly so crowded in those days as it is now. It was quite easy to sit undisturbed at a table for a whole morning.

Kuhn:

Do you have any recollection of how your time was distributed not between libraries, but between reading the literature, working on problems yourself, working up subjects that you were bearing lectures about, or actually going to lectures?

Dirac:

I went to certain courses of lectures, and did a good deal of reading on my own. I get more precise information from my own reading than from lectures. I don't seem to take it all in when I go to a lecture; I don't get the chance to refer back in the way I want to understand the thing properly. So I just get general ideas from lectures. The rest of the time I was just reading papers and thinking about them and trying to improve on them.

Kuhn:

When you read to work up a subject, do you regularly try, simultaneously, to work the things out yourself? I mean, do you read with a pencil and paper, and do at least go through some of the things?

Dirac:

If I really want to understand it thoroughly, I usually put it into my own notation, because other people's notation is usually not very suitable; it's unnecessarily awkward, and I like to put things in my own notation and get everything as simple as possible.

Kuhn:

Did you think you did that with most of the subjects you were listening to lectures on in that first year?

Dirac:

Probably not with most of them, but with the things that I was mostly interested in. The Hamiltonian thing I worked on a lot, by myself, and I used the Whittaker book also, Whittaker's Dynamics. In those days one thought that the action and angle variables were the all-important things. They come in to the Bohr Sommerfeld quantization. The question was to try to introduce action and angle variables for motion which was not periodic, or to introduce something corresponding to them.

Kuhn:

When you say here, 'not periodic' are you thinking of the generalization from periodic to multiply periodic?

Dirac

Yes.

Kuhn:

Did you worry at all in these years about doing something for the problems which were not even multiply periodic?

Dirac:

Well, I understood that they ought to be brought in, but I didn't know any way of doing it. I thought a great deal about it without any success.

Kuhn:

That sort of information, Sir, where it can be recaptured — you know, things that you were working on and didn't succeed with, but which seemed the real problems whether they got anywhere or not — is terribly useful, and I think quite important. It's particularly important for this sort of work because it's just the information that, except where large amounts of notes are available, would not be forthcoming any other way. One must try to discover the shape of the problem structure of the field in the years just before the new quantum mechanics emerged.

Dirac:

I know I was very much impressed by action and angle variables. Far too much of the scope of my work was really there; it was much too limited. I see now that it was a mistake; just thinking of action and angle variables one would never have gotten on to the new mechanics. So without Heisenberg and Schrödinger I should never have done it by myself.

Kuhn:

You build action and angle variables, or something like them, also very deeply into your own early work in the new mechanics.

Dirac:

Yes, yes. Of course, I was then trying to fit in the new mechanics with my previous ideas of action and angle variables.

Kuhn:

Where does the term 'uniformizing variables' come from? You use this I think repeatedly, and I don't think I've seen it elsewhere, but I may have.

Dirac:

I can't remember; maybe Fowler used it in his lectures. I know I was very much under the influence of Fowler and used the same sort of language that he used.

Kuhn:

Did you see a great deal of him from the time you first arrived?

Dirac:

Yes, yes. He was officially my supervisor; that meant he was responsible for my work, and I'd go and see him maybe once a week, or perhaps not every week. But roughly once a week.

Kuhn:

For how long would you be with him? And what would go on in these sessions?

Dirac:

Well, I'd just tell him about my work, and talk things over with him. I remember having one argument with him about something in the variation method, I didn't agree with what was put in the books, and Fowler thought that what was in the books was quite right, and he got a bit impatient with me. But I couldn't see his point; I think perhaps that early difficulty of mine was what led later on to constraints in the Hamiltonian theory. That was much later, you see.

Kuhn:

This is in quantum mechanical —

Dirac:

Well, it's really in classical mechanics, also. My generalization of Hamiltonian dynamics which I worked on around 1949 and so on. It was very much later, but I think this was really straightening out that early difficulty.

Kuhn:

I don't know that piece of work at all. The older classical theory of constraints I once learned.

Dirac:

This is another kind of constraint.

Kuhn:

I would be glad for sort of a greater sense of the extent of Fowler's activity in guiding your reading and in talking with you about problems.

Dirac:

I think there was only that one occasion when I had a disagreement with him, and he got impatient with me.

Kuhn:

Did he generally follow quite closely what you were doing, or were these sessions more likely to be just 'checking in'?

Dirac:

I think he followed it and understood it, oh yes.

Kuhn:

Did he also, at the beginning, guide your reading a good deal, or did you pretty much pick that yourself?

Dirac

In the beginning he would tell me what to read. In one of your questions you asked whether I liked writing up papers; well, I didn't like it at all. When I first had any work to write up, and I told Fowler I didn't like writing up, he said, "Well, if you're not going to write your work up, you might as well shut up shop." He put it as definitely as that, and I knew it was just something that I had to force myself to do.

Kuhn:

Tell me more about writing up papers. Did. you do many drafts?

Dirac:

Not very many drafts. Words don't come to me very easily; I first have very rough notes, and perhaps two or three drafts, not an awfully lot. Not at all like what Bohr did.

Kuhn:

Is the paper pretty much done in your mind before you start to write it, or do you find other things in the course of writing, of trying to express it and react back and modify?

Dirac:

Well, I think it's pretty well done, and in the course of writing it up I may find I need serious alterations.

Kuhn:

Is it usual to find that serious alterations are needed?

Dirac:

Fairly often. I don't know whether it would be less than half, or not probably less than half,

Kuhn:

I know in my own work I've very often found that, but I've never been quite clear how common it was.

Dirac:

It's really quite irregular; one can't generalize about that.

Kuhn:

Do you show these drafts to people?

Dirac:

Not as a rule, no.

Kuhn:

So that you write it, rewrite it, rewrite it and then submit it?

Dirac:

Yes, yes. I don't actually rewrite it so much as make a lot of corrections in it. When I don't like writing, I try to cut that down to a minimum. I use an eraser very much.

Kuhn:

So it's a matter of a draft, many corrections, and then a final clean copy?

Dirac:

Yes, yes.

Kuhn:

I do notice, probably more in your papers than in any others that I have read recently — I think this is perhaps truest of your early papers a tendency, one or two papers later, to come back in the introductory section with a fairly substantial sketch of a method that has been laid out in an earlier paper, but which you're going to use again.

Dirac:

That is when the earlier method has been improved on. I think it is usually just when the earlier method has been improved upon. I think I've got an example of an early draft here. (Goes to look for it) It's in one of these papers I noticed

Kuhn:

I would have thought, and we may be able to explore this more clearly when the draft shows up, that in at least some of these cases, it really was not a matter of an improvement in the method, at least at the most technical level. The formulas might all be the same, but what seems to come through is a clearer and clearer exposition of what it's about, of why one does things this way, or just what is represented in

Dirac:

Very often it was just a question of putting the ideas in the right order, so that they can be best absorbed by the reader.

Kuhn:

When that happens do you think it was usually a matter of your own realization that the presentation might need clarification, or did people come to you and say, "I don't understand what you mean here"?

Dirac:

It was usually by myself, just thinking of a logical way of setting this thing down for someone who doesn't have previous knowledge of the subject.

Kuhn:

Were you from the beginning concerned deeply with clarity in the presentation; with the problem of the audience? People seem to vary tremendously in this respect; some people say, "If I get it down they should try to figure out what I mean."

Dirac:

I was more concerned with getting it clear in my own mind explaining it to other people was secondary to that, but when I did have it clear in my own mind then of course it wasn't so much trouble. It was just rather tedious to write the things down.

Kuhn:

Besides Fowler, were there other people that you saw with any regularity, either staff or

students, in this first year or two in Cambridge?

Dirac:

Well, there were some that I saw frequently, but I wouldn't say regularly. There was Milne for instance; I went to lectures by him on astrophysical problems. Eddington I saw occasionally, but not very often.

Kuhn:

Did you see Milne outside of lecture as well as in?

Dirac:

Well, I did during the period when he was my supervisor, but I'm not sure about other times. There were the meetings of the various clubs where I would meet people.

Kuhn:

But on the whole, there was no one you think of as somebody with whom you really talked over problems, unless it was with your supervisor?

Dirac:

That would be the only person, yes.

Kuhn:

That would be equally true of other students?

Dirac:

Yes. I don't think it applies so much nowadays because so many people are working on closely related subjects, and they can get together very much.

Kuhn:

I get the impression that Fowler was really, until your own time, perhaps the only one here at Cambridge who was very much concerned with the problems of quantum physics and the only one from whom people would have learned about them. Is that fair?

Dirac:

Well, there was Rutherford on the experimental side. Rutherford and Fowler were a team; they dominated the whole field of quantum theory. Lennard-Jones to some extent - she was called Jones in those days; she became Lennard-Jones when she got married.

Kuhn:

One thing I raised in the questionnaire, but again a thing I would perhaps know better simply if I knew Cambridge or the British University system, is this for me rather unfamiliar division of what I think of as the physics curriculum into applied math and experimental physics, natural philosophy.

Dirac:

It's a division between those who do theoretical work and those who do experimental work. All those who do theoretical work are counted as in the mathematics faculty.

Kuhn:

Now, there is of course in the United States often an appreciable division between those who do theory and those who do experimental work, but it's rather in spite of the curriculum than because of it. They're in the same department, the examining requirements are likely to be much the same for the two, although there will be enough flexibility so that the subjects offered will often be different. What I wonder then really is, how separate were these two groups, how much interaction was there with the experimentalists? How different was the preparation?

Dirac:

There was quite a lot of interaction during the days of Fowler and Rutherford. From the point of view of administration we had a close connection between pure and applied mathematics; we tried to keep students doing both pure and applied to quite a late stage and only specializing between them perhaps in the third year. In keeping pure and applied mathematics together it meant of course that the applied mathematics was more detached from physics. But Fowler and Rutherford were very close together. You know that Fowler married Rutherford's daughter?

Kuhn:

Oh yes. Did that closeness affect the students also?

Dirac:

Oh yes. I mean we would go to experimental colloquia: and there were the clubs which

were divided between theory and experiment. They tried to keep the “Del Squared V” fifty-fifty between theoretical and experimental people. Also there would be quite a lot of contact between theoretical and experimental students through these clubs.

Kuhn:

What about the colloquia at the Cavendish?

Dirac:

I always went to them.

Kuhn:

Would they be pretty much exclusively experimental?

Dirac:

Mainly experimental, not always; I have talked there sometimes myself. And Bohr has talked there; that’s another meeting place for the theoretical and experimental physicists.

Kuhn:

... The impression I’m getting is that this administrative division between physics and mathematics did not prevent, in your period at least, a quite close interrelationship between the experimentalists and the theoretical people.

Dirac:

No. I suppose the separation applied mainly to the undergraduates, because there would be a quite sharp division in the courses according to whether they were theoretical or experimental.

Kuhn:

Can you tell me more about what that would have amounted to? How much mathematics would somebody who was going to be an experimental physicist get?

Dirac:

Well, he took calculus and Maxwell’s equations and I don’t know where he would go beyond that.

Kuhn:

Would he have a mathematics examination?

Dirac:

I think he would, yes. I'm not very sure about this. I don't know whether you're asking about these old days, or the present.

Kuhn:

Well, on the whole I'm asking about the old days. You, I take it, did not take any Cambridge examinations as a research student, so that this issue of preparation for exams didn't arise with you.

Dirac:

No. It didn't arise with me at all. Of course I found that I knew less than if I had taken the Cambridge examinations; the mathematics examination in Cambridge was, well, a good deal more advanced than what I'd had in Bristol.

Kuhn:

Was there any particular thing that you discovered you were missing? Was it just that the level of the work was more advanced, or were there whole subject matters gone?

Dirac:

I think the level was more advanced, and there were whole subjects, like thermodynamics which were quite new to me. And then Gibbs statistical mechanics — that was quite new to me. There were whole subjects that were quite new.

Kuhn:

You have mentioned on more than one occasion having made great use of Sommerfeld's when you started to work with Fowler and learned about atomic theory and quantum mechanics.

Dirac:

Yes.

Kuhn:

Were there other things besides that that played a particular role? I wonder, for example, about Bohr's 1918 Quantum Theory of Line Spectra which was a hand book for some people? Or a little later -

Dirac:

I didn't study it very much, if at all.

Kuhn:

Do you remember whether you ever studied in any detail the Kramer's paper on the hydrogen atom which came immediately after that?

Dirac:

Where was that published?

Kuhn:

It is published in the Danish Academy Proceedings, as Bohr's was, I think just in the next year.

Dirac:

I don't remember it. I probably read most of the papers that were concerned with quantum theory; there were not nearly so many papers then as there are now. It was not so difficult to keep abreast of them.

Kuhn:

No, these were actually of course things that had come out before. These came out in 1918 and 1919.

Dirac:

Yes, well I had to go back to those.

Kuhn:

But you think of Sommerfeld's as having been perhaps the main book to which you returned and worked with?

Dirac:

That was the only book I think, and the rest was individual papers.

Kuhn:

Some of them get to be almost book length, that is, if you put the three Bohr parts together. What about Born's Atommechanik? This is a bit later; it's before you do your thesis.

Dirac:

I don't remember; was that already on quantum mechanics?

Kuhn:

It's one of the last book length works on the old quantum theory; in fact it's the book that he calls volume 1, saying at the beginning that it's so clear that this whole thing has to change that 'I hope to be able to write a volume 2 soon that will reformulate everything.'

Dirac:

I think it came out rather too late by the time I got to it; the new quantum theory had already appeared. I remember that book; I didn't really read it very much.

Kuhn:

I have been much concerned myself with trying to discover how different institutions differed in the period from '20 to '25, say, in their awareness that really fundamental changes were needed, and also in their consciousness, which also sometimes goes with this same distinction, of how pressing the old paradoxes were that existed almost since Planck or in some cases right since Planck. Such paradoxes as the conflict between classical theory and quantum postulates, wave particle dualism which comes later, the sort of almost cosmologic contradiction on the one hand; and on the other hand there are those whole series of current problems that aren't quite being solved, like the helium atom, and the more general problem where you have the more general center of force. There were also more technical problems that may just work out, but at least in Copenhagen and Göttingen by 1923 or thereabouts people are quite clear that these things aren't going to work out without a basic reformulation.

Dirac:

I think they were not so clear in Cambridge. I was always thinking it would be in terms

of action and angle variables.

Kuhn:

As we said yesterday, you are the first person to see and to state really clearly just where Heisenberg has broken with classical theory, pinning it right down to non-commuting variables.

Dirac:

Yes, well you told me that Heisenberg also appreciated that point very early.

Kuhn:

Well, he appreciated it as a problem; he thought that it shouldn't be • He was bothered by the fact that the variables didn't commute. You, on the contrary, point out that except at that one point, there is a full parallelism here and we can therefore take everything out of classical mechanics, break it at this one point — which of course means that there are certain sorts of classical derivations which we can't do the old way — but otherwise preserve a full parallelism.

Dirac:

I think there are many examples where the person who first introduces a new idea is bothered by something which doesn't agree with the old work, while other people just seize on it as the important thing. I can give one example of that which came later with the negative energy electrons. I felt right at the start that the negative energy electrons would have the same rest mass as the ordinary electrons, and that bothered me very much. I felt that such positrons like that could not exist; otherwise the experimental people would have discovered them. That was my main worry at that time; I hoped that there was some lack of symmetry somewhere which would bring in the extra mass for the positively charged ones. Weyl was the first to point out quite definitely that the holes would have to have the same rest mass as the electrons. He wasn't bothered by it the way that I was.

Kuhn:

This is of course way ahead, but I don't want to let go of it. How early was that bother? I mean, I notice one thing particularly: the wave equation is '28; it's almost two years later before you come out with the proton theory, and I have no notion how, during the intervening two years, that problem has developed. You're quite explicit in the papers on the wave equation that the problem exists, not of course in terms of an expression in hole theory; but rather that there is this old problem of the negative energy levels which

is a classical as well as a quantum mechanical problem, only you can't discard the negative energy levels in quantum mechanics. In these papers, or this pair of papers, however, you are concerned only with the other part of the problem of relativistic theory. Then you just drop the question of the negative energy solution, so far as the published work is concerned, for almost two years.

Dirac:

It was an imperfection in the theory; it bothered me very much and I didn't see what could be done about it. It was only later that I got the idea of filling up all the negative energy states.

Kuhn:

Had you been coming back to that problem repeatedly over the intervening period?

Dirac:

Well, it always had been in my mind.

Kuhn:

But presumably the notion of filling up the other states and treating these solutions as holes didn't come to you until fairly shortly before you actually worked up the paper?

Dirac:

That is so, yes. Then I was bothered by the rest mass being the same.

Kuhn:

My impression is that in the actual paper, you don't really deal with that problem, with the lack of symmetry, in the 1930 paper.

Dirac:

Well, I think I say I hope that there is something which brings in a dissymmetry. I think I said that I hoped that the Coulombian interaction would bring in that dissymmetry. I felt then that if it did not bring in the dissymmetry, the whole theory would have to be counted as wrong.

Kuhn:

Well, now let me bring you back to this question that we were getting at when I raised the question of isolation — the nature of Heisenberg's break. Also, I think you felt that he was preserving much more than I think he realized he was preserving. Your whole clarification, from the very start, of the relation between Heisenberg's variables and the classical variables I think makes it look vastly more classical than it has in Heisenberg, and simultaneously isolates cleanly the single point at which you say that he has broken with the classical thing. I raised this because I wanted to ask you to what extent all that you had been doing and thinking about yourself, had prepared you for a break of this sort. Not that particular break, but for something that was as unclassical in a sense, or unHamiltonian.

Dirac:

I think it hadn't prepared me at all, and it was quite a surprise to me. I could say something in further elaboration of the previous point. A person first gets a new idea and he wonders very much whether this idea will be right or wrong. He is very anxious about it, and any feature in the new idea which differs from the old established ideas is a source of anxiety to him. Whereas someone else who hears about this work and takes it up doesn't have the same anxiety, an anxiety to preserve the correctness of the basic idea at all costs, and without having this anxiety he is not so disturbed by the contradiction and is able to face up to it and see what it really means. I expect that was just Heisenberg's problem. He was afraid that this lack of commutation might cause the whole theory to collapse; he was probably terribly worried about that. That rather stops him from really facing up to it.

Kuhn:

I hope you will continue to say things of this sort also as we go, because they're immensely helpful.

Dirac:

I think that applies quite generally to all new ideas which are brought in by anybody.

Kuhn:

Can you think of other cases of that sort, either involving yourself, or involving others?

Dirac:

I expect if I thought about it, I would remember some examples. They don't occur to my mind immediately.

Kuhn:

It would be terribly illuminating to pick up more examples of this [sort of thing] from this period, and particularly from your own work if you think of them, on the same point of the sense of crisis and the sense of the need for a break from Copenhagen in particular, but also to some extent from Göttingen in the period from '23 to '25.

Dirac:

Well, I do think of one example, namely, the relativistic theory of the electron. When I first got that equation, of course I was very anxious to know whether it would work for the hydrogen atom, and I just tried it by an approximation method. I thought that if I got it anywhere near right with an approximation method, I would be very happy about that. It needed someone else, namely Darwin, to tackle that equation as an exact equation and see what the exact solutions were; I think I would have been too scared myself to consider it exactly. I would be too scared that it would get unfortunate results which would compel the whole theory to be abandoned.

Kuhn:

That's fascinating; does this mean that you had yourself not tried to handle it exactly before going to an approximation method?

Dirac:

That is so, yes.

Kuhn:

You looked for the approximation method from the start?

Dirac:

Yes, yes. Of course I had the fear that the whole theory was nowhere near right, and if I could get it approximately right, well, my confidence would already be substantially increased in that way. It's just that one has lack of confidence when one introduces something quite new.

Kuhn:

That's terribly interesting. I will ask you at this point a question that I had intended, to ask later. When you got the equation, were you looking for an equation which would give spin, or were you looking for a relativistic equation?

Dirac:

I was looking for a relativistic equation.

Kuhn:

When you did the approximate solution, was what you were hoping to approximate something with or without spin? I mean was it a surprise that what came out were spin terms?

Dirac:

No, I don't think so, because one had the Pauli matrices in it. I remember when I was in Copenhagen quite early Bohr asked me what I was working on, and I told him I was trying to get the relativistic theory of the electron. And Bohr said, "But Klein has already solved that problem." I was rather perturbed by that, but Bohr seemed to be quite complacent and satisfied with Klein's solution, and I wasn't. I remember it disturbed me quite a lot that Bohr was so satisfied with it because of the negative probabilities that it led to. I just kept on with it,

Kuhn:

At what point did your insistence on a linear equation and the preservation of transformation theory enter your work on that problem?

Dirac:

Well, previously the transformation theory had been set up in a general form and I felt that that was correct and had to be preserved, and had to be fitted in with relativity.

Kuhn:

So that was really the basic guideline of the attempt, to get a relativistic formula from the beginning?

Dirac:

Yes. It was just that I had confidence in the transformation theory of quantum mechanics. I suppose one's research is guided very much by what one has confidence in and what one is feeling doubtful about. But I suppose I had more confidence in that transformation theory than other people did, I felt it imperative to keep to positive probabilities.

Kuhn:

Either someone told us or in one of the accounts of things going on in Copenhagen it is written down that the problem that you raise at the beginning of the paper on the wave equation, “Why isn’t Nature satisfied with a point particle, why spin?” was also very much on your mind in Copenhagen. Possibly not on your first trip to Copenhagen, but it is said that at least it was a problem that you had raised and discussed there.

Dirac:

Not to the same extent. The main thing was to extend the transformation theory of quantum mechanics.

Kuhn:

Do you remember at what point you suddenly saw spin coming out — saw that the extra variables that you were introducing?

Dirac:

Well, one had it non-relativistically before then. It came out just from playing with the equations rather than trying to introduce the right physical ideas. A great deal of my work is just playing with equations and see in what they give. Second quantization I know came out from playing with equations. I don’t suppose that applies so much to other physicists; I think it’s a peculiarity of myself that I like to play about with equations, just looking for beautiful mathematical relations which maybe don’t have any physical meaning at all. Sometimes they do.

Kuhn:

Well, now, in the case of the wave equation, what do you suppose you were here playing with - a general linear form?

Dirac:

Yes, yes.

Kuhn:

And it’s somewhere then in the manipulation of the alphas that you begin to discover spin, that you can do things with the Pauli spin matrices, or that you get conditions that look, in some cases, like the Pauli spin matrix conditions?

Dirac:

I remember noticing that just forming the three dimensional scalar product of sigma with the momentum gives you something which is quite nice to play with, and I wanted to extend that to four dimensions. And of course one just can't do it sticking to the two by two matrices, and it needed quite an effort to make the further generalization to the four by four matrices. But that work did come from playing about with the three dimensional scalar product and trying to extend it.

Kuhn:

How long do you suppose that went on?

Dirac:

Maybe well, it was a matter of weeks, not more. I don't know whether you knew that Kramer's had independently discovered the second order wave equation which you can get by multiplying up my first order wave equation. He told me afterwards that that second order equation he had found himself. I don't know how he found it, but just by trying to bring the spin into relativistic theory, I guess.

Kuhn:

No, I had no idea of that. That I take it was not published?

Dirac:

No, it was not published. I suppose he was just working on that sort of question when my paper came out, and then it was not necessary for him to continue that line of work.

Kuhn:

As I understand this line of development you were not particularly working for spin, but that in fooling around with the equations, playing with the linear form, you began to see things like the Pauli spin matrices and began to see that spin might, and ultimately was, going to come out of this equation into which nothing of the sort had gone in. My question then really is, can you remember what it felt like, to see this really quite unanticipated result two puzzles tying together this way?

Dirac:

Well, in the first place it leads to great anxiety as to whether it's going to be correct or not. That anxiety that I told you about before. I expect that's the dominating feeling. It gets to be rather a fever; you work things out with it and see whether it's going to turn

out right or not.

Kuhn:

When that fever begins to develop and the anxiety builds up, do your working habits change accordingly? Do you find yourself working much longer hours? Do you have trouble sleeping? Do you take more walks, or fewer walks?

Dirac:

Well, I take less interest in the outside world. But the excitement does prevent one from working things out in an ordered way. If one was not so emotionally excited by it, one could get on faster.

Kuhn:

In the '23 to '25 period, as you yourself were getting quite deeply involved with quantum mechanics, there's a lot of work going on in Copenhagen and secondarily in Gottingen on extensions of the old quantum mechanics the whole notion of the non-mechanical forces and the attachment of two quantum numbers to coupling. The Bohr-Kramer's-Slater paper, and the non-conservation of energy in the virtual oscillators, the Kramer's dispersion formula, are all products of it. And of course, Heisenberg's paper, in a sense, is also a product of this great broadening out of classical Bohr quantum mechanics. I wondered how much of that work was really known and followed closely. And this is work done by people who are pretty clear that there is going to have to be a major change in a way that, I take it from what you say, you were not really clear it would have to change.

Dirac:

I think that is so, yes.

Kuhn:

Were you much aware of that work? There's a paper of Heisenberg's with three different ways of —

Dirac:

I was aware of the general ideas, but I didn't follow all the details.

Kuhn:

Were you at all sympathetic to them?

Dirac:

Well, when the Bohr-Kramer's-Slater idea came out I thought probably it was right, to begin with. I tried to adjust myself to it, but of course it didn't survive very long.

Kuhn:

Well, the non-conservation of energy' didn't survive very long; the virtual oscillator approach, though, goes straight into the Kramer's' dispersion formula and the Kramer's-Heisenberg paper and into a number of other broadenings of the Correspondence Principle. And in that sense it goes into Heisenberg's matrix mechanics without matrices' paper.

Dirac:

I don't think I was very strongly influenced by this work. I have a general view that it is best to let people work out their own ideas.

Kuhn:

What about the Kramer's dispersion formula?

Dirac:

Well, I was interested in it, but I didn't see how it could go on from that. When one reads about work like this the question arises, "Is it a starting point from which one can go on?" and I didn't see how to go on from that dispersion paper. But from the Heisenberg matrix paper, well, I did see how to go on from that.

Kuhn:

In your own paper, in which you go on from the Heisenberg paper, you refer to the Kramer's-Heisenberg dispersion paper as giving an example of a case in which the derivative of one q-number with respect to another comes out in a form that can be looked upon as a commutation form.

Dirac:

Well, I don't remember that. Do you want me to refer to it?

Kuhn:

Yes. This is the paper on the fundamental equations in quantum mechanics. [Paper No. 8] [Dirac searches for the paper, reading some other titles]

Dirac:

Here's that —. That's this "Adiabatic Invariance of the Quantum Integrals." [No. 5] I think that must be Fowler's comment up there. [referring apparently to marginalia]

Kuhn:

This is really an extremely condensed version isn't it, of the —?

Dirac:

Yes, I've quite forgotten what it is; I haven't read it (for a long time).

Kuhn:

Well, actually I notice that this is an outline of the method. 'The first two formulas that you introduce are actually numbered and 6. Have you many drafts that go back to this period?

Dirac:

I didn't know I had any; I just happened to see that the other day. There might be a few more if I searched.

Kuhn:

I was thinking particularly of this paragraph here which goes back to - well, your revision goes back to 9, I see, that actually makes a substantial difference. It may throw off my whole question.

Dirac:

What was the question?

Kuhn:

Well, in the printed work here, this is the reference to the Kramer's-Heisenberg theory. [i.e. in. connection with equation 8: $dx/dV = x_a - ax$]

Dirac:

There is a misprint; I wonder if that helps any.

Kuhn:

Well, it surely does, because 8 was this, which is a very fundamental relationship. I didn't in fact understand, but I haven't gone back and tried to piece together a way in which one might put the Kramer's-Heisenberg formula that way. If it's 9 it may still be a question, but it becomes a different question.

Dirac:

Yes, I suppose there is some comment which probably is made later on.

Kuhn:

I might, having gotten this out, ask you one other question. I may be missing a point in the nature of the argument in this paper, but I'm perplexed here about the way in which the whole question of the derivative of one q-number with respect to another enters in this paper • The third section is called quantum differentiation, and comes very neatly to the conclusion. that it's got to take the form of a commutation relation. On the other hand, that's the only place in which I can see that the question of differentiation really enters here. You could, I think, have gone right on to the question of the quantum conditions in paragraph 1 and right through the rest of the paper without ever explicitly introducing quantum differentiation at all. That suggests to me that it plays an essential role in the way you get to this.

Dirac:

Yes, I think I was looking for something to replace the partial differential equations of the Hamiltonian theory.

Kuhn:

And perhaps you were looking for that before you thought of writing them in the Poisson bracket form directly?

Dirac:

Yes. I know I was working for quite a long while on trying to get a connection between the Heisenberg formulas and the action and angle integrals, and the Hamiltonian equations of motion.

Kuhn:

And this was written not in the form of Poisson brackets, but in the more usual partial differential form?

Dirac:

Yes, yes. In the usual differential form.

Kuhn:

That would exactly answer the question.

Dirac:

I didn't know much about Poisson brackets at that time. Did I tell you how I first came to think of the Poisson brackets?

Kuhn:

No. You told me in part when, but I'm very much interested.

Dirac:

I used to take long walks on Sundays and get away from the work altogether, and at the end of those walks I would perhaps go on with my work a bit in a refreshed state of mind. And after one of these Sunday walks it occurred to me that the commutator might be the analogue of the Poisson bracket, but I didn't know very well what a Poisson bracket was then. I had just read a bit about it and forgotten most of what I had read, and I wanted to check up on this idea, but I couldn't do it because I didn't have any book at home which gave Poisson brackets and all the libraries were closed. So I just had to wait impatiently until Monday morning when the libraries were open to go and check up on what Poisson brackets really were. Then I found that they did fit, but I had one impatient night of waiting.

Kuhn:

I'm wondering about the preparation before this walk. You said when Fowler had first given you the proof of Heisenberg's paper, you had looked at it and hadn't thought it amounted to much.

Dirac:

That is so, yes. I don't remember what my earlier reaction was; I've often tried to recall it, but I can't remember what it was. I supposed it was just some disparity between that and the Hamiltonian formalism. I was so impressed then with the need of Hamiltonian formalism as the basis of atomic physics, and anything that didn't connect with it I thought wouldn't be much good. I expect it was something on those lines, but I don't remember just what it was.

Kuhn:

Do you have any notion what it is that brought you back to look at it again, or what it was that you began to see in it when you began to pay more attention to it? I take it, namely, that there is a stage that intervenes between this putting it aside and the decisive walk, when it occurred to you that the Poisson bracket might be the analogue; but you've been working quite hard. in the attempt to bring it into a Hamiltonian form meanwhile.

Dirac:

I first got the paper in September, 1925, the beginning of September, and I think it was perhaps the middle of September when I went back to it again. I suppose I realized that it was introducing the new idea of the non-commutation. It was already in September, 1925, that I realized that this paper did, mean a breakthrough.

Kuhn:

You don't really remember what it was about it that suddenly made it seem more important, or that convinced you that perhaps this was a break-through?

Dirac:

No, I don't. In fact it seemed pretty obvious then, and I really find it more difficult to understand why I didn't see it the first time I looked at this paper. But I then worked quite hard to try and connect these matrices with the action and angle variables, and that took a good many weeks, maybe a month or two, before this walk occurred in which I got the idea of the Poisson bracket.

Kuhn:

Did you try to get any applications into the first paper? That is the first paper cuts off very sharply with just a presentation of the formalism, if you will.

Dirac:

I expect what happened was that I showed it to Fowler, and he was very much

impressed by it and saw the need to publish it urgently. He told the Royal Society to give priority to my paper, and they published it extremely quickly. I expect you notice that with these early papers of mine there's only a very short interval of time between when the paper was sent in and when it was published. Well, I can thank Fowler for that, because he appreciated that it was urgent, and I suppose he was thinking that there would be competition from other places. I expect Fowler told me that I ought to publish what I had, and then go on to further papers from there.

Kuhn:

In the papers that immediately follow this one, particularly then the work on the hydrogen atom and then on the more general multi-electron atom, you effect a gigantic versatility with q-numbers. You know, one can follow your arguments, but one can't imagine having invented them. How easy did that —

Dirac:

I think it came pretty easily; I can't remember having any special difficulty with it. The difficulty was with the physical interpretation, and I think it took a year or two to get that straightened out.

Kuhn:

But the problems of the approximations and finding the transformations and so on, came fairly easily and directly? There is some awfully elaborate q-number algebra, or q-number geometry - we'll leave open the question of which — in the paper on the elimination of the nodes. [No.10] I've read through them, but in those cases I certainly have not tried to go through the entire -

Dirac:

I'd call it algebra; I could settle down to algebra when I had the basic ideas given, but to get new basic ideas I worked geometrically. I think perhaps that clears up that discrepancy. For getting new ideas I worked geometrically; once the ideas are established, one can put them in algebraic form and one can proceed to deduce their consequences. That's just a question of algebra. The more difficult part, and the more important part, is the getting of the new ideas, and that requires a geometrical mind, I believe. I suppose that it's to some extent the geometrical mind when you think that q-numbers can be used with a great deal of analogy to ordinary numbers. I suppose that was the main point in my early work, that I did appreciate that there would be a very close analogy between the q-numbers and ordinary numbers.

Kuhn:

Your approach through q-numbers really, from the first paper after Heisenberg's on, is, as you yourself point out, rather different from the Heisenberg-Born Jordan approach through representations and through the matrices, taking the matrices to be themselves fundamental.

Dirac:

Yes • I see now that that comes just from realizing that the most important thing is the non-commutation, just the thing which disturbed Heisenberg.

Kuhn:

But here, on that point also, Born-Jordan are also quite explicit and pull the non-commutation out as at least also their form of the —

Dirac:

They don't have the same idea as Heisenberg would have about it.

Kuhn:

You pushed for some time, and with immense, fruitfulness, a rather different way of setting up and getting at quantum mechanics from the matrix mechanics way.

Dirac:

Yes, What one might call a symbolic way.

Kuhn:

With representations to be secondary. Certainly to me it has that sense of being a more fundamental approach. It also raises some problems, since you've still got the representation to get out when you've this.

Dirac:

Yes.

Kuhn:

I wonder what sort of problems this may have made at the time, for you, and for other people trying to read the works and put the various pieces together. Did anybody ever urge you to get on the matrix bandwagon?

Dirac:

No, no. Fowler was certainly quite happy that I was getting on my own wagon. He certainly wouldn't urge me to change my method at all.

Kuhn:

Was it ever tempting? Did you ever think seriously —

Dirac:

No if you've got a good method, you stick to it. And anyway my method gave results quickly, without requiring very much writing; I'm a (lazy) person who doesn't like to write things up.

Kuhn:

If one can do that much in one's head. Wouldn't it be fair to say that an awful lot of other people have been able, with great labor, to get results out of, let's say, the matrix method, which is in a sense fairly straight-forward. You can write it all down, hold all the indices, finally solve all the differential equations. But there isn't any sort of parallel mechanical way to do the q-number manipulations, so that if you can't see them in your head and therefore do them quickly, you're going to have great trouble doing them at all?

Dirac:

I don't think that it's quite right to say that I see them in my head; I have to play about a good deal with these q-numbers and see what relations they give. I find that some of these relations are useful, and I try to get the useful relations in the simplest possible way, and then I publish that • And all the intermediate steps get scrapped. So it wasn't a question of finding out these methods straightaway. They were often obtained only by very circuitous —. I have a sort of recollection that the work that I did on the hydrogen atom with the spinning electron equation was very much on those lines. And the method for handling the angle variables in that problem I put in the published paper in the simplest possible form, but that was a very different form from the one in which I first worked it out. You must first work it out and get the results and then labor on that quite a lot until you see how to get the form in the most direct way. This kind of procedure is what is done by Marcel Rises quite a lot; I don't know whether you know him.

Kuhn:

No.

Dirac:

He's a pure mathematician; he's really very much concerned with putting his results in a tidy and beautiful form, and he really considers that the main part of his work is not to obtain the results, but after you've obtained them to find out the neatest and most direct way of getting the results. I didn't meet Rises until very many years later, but I had been following the same sort of method he follows. He worked for a long time in Lund, but he has retired now.

Kuhn:

You said before that the algebra came fairly easily, that what was tough and took a long time was the interpretation. Were you concerned with the interpretation from the very beginning?

Dirac:

Oh, yes, yes; it's a physical theory, so one had. to have some interpretation for the q -numbers.

Kuhn:

In talking to Heisenberg and to a lesser extent talking with Born, I raised this same question. You know that whole series of matrix mechanics papers is notably free from any questions about, "What's this all about and how are we to understand it physically?"

Dirac:

Well, I shouldn't have thought there was that problem in the matrix formulation, because you have the matrix elements with their direct application.

Kuhn:

That does answer it. It's a much more minimal sort of answer, isn't it, than the one that emerges, but I see how it stops the question.

Dirac:

The whole idea of the matrices was to work entirely with numbers which have a physical interpretation.

Kuhn:

It can leave all sorts of questions which emerge immediately. The problem that you're faced with with the q-number which has not yet been represented by a matrix is a question that you can also be faced with by the matrix if you ask not, 'what does each of these elements mean?' but 'what does the array mean — why are they all there?'

Dirac:

Yes, you don't have any problem with the interpretation then.

Kuhn:

No.